

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q19-1298

Jarales Tap 138 kV
August 23, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1298. The Interconnection Customer proposes a 235.8 MW solar photovoltaic facility interconnecting at the Jarales Tap 138 kV substation in the Pennsylvania area. The request was received on September 27, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 351 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 6 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$44,285,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

The HUMBOLDT BAY-WATSONVILLE 138kV line is overloaded to 133.9% of its 158 MVA emergency rating for the loss of the TEXACO TRADING-JARALEs TAP 138kV. The project contributes 100.0% DFAX (235.8 MW) to this overload.

HUMBOLDT BAY-HELICON 138kV: 118.7% loading (184 MVA) following the outage of the TEXACO TRADING-JARALEs TAP SUBSTATION 138kV; DFAX = 100.0%.

The HELICON-COMO 138kV L is overloaded to 115.0% of its 193 MVA emergency rating for the loss of the TEXACO TRADING-JARALEs TAP 138kV. The project contributes 100.0% DFAX (235.8 MW) to this overload.

The COMO-JARALEs TAP 138kV is overloaded to 111.3% of its 206 MVA emergency rating for the loss of the TEXACO TRADING-JARALEs TAP SUBSTATION 138kV line. The project contributes 100.0% DFAX (235.8 MW) to this overload.

The TEXACO TRADING-JARALEs TAP SUBSTATION 138kV is overloaded to 106.6% of its 215 MVA emergency rating for the loss of the COMO-JARALEs TAP 138kV. The project contributes 100.0% DFAX (235.8 MW) to this overload.

PASTILLO 5801 138-1 kV TX: 103.0% loading (319 MVA) following the outage of the TALENENERGY MARTINS CREEK LLC WILLIAMSPORT-MONTOUR STA 345kV; DFAX = 48.4%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
WATSONVILLE-HUMBOLDT BAY 138kV	Rebuild line with higher-capacity conductor	\$7,242,000	100% (\$7,242,000)
HUMBOLDT BAY-HELICON 138kV line	Rebuild line with higher-capacity conductor	\$5,276,000	100% (\$5,276,000)
HELICON-COMO 138kV L	Rebuild line with higher-capacity conductor	\$7,706,000	100% (\$7,706,000)
COMO-JARALES TAP 138kV	Rebuild line with higher-capacity conductor	\$12,917,000	100% (\$12,917,000)
TEXACO TRADING-JARALES TAP SUBSTATION 138kV	Reconductor line	\$5,354,000	100% (\$5,354,000)
PASTILLO 5801 138-1 kV TX	Replace transformer with higher-rated unit	\$6,087,000	95% (\$5,790,000)
	Total allocated to project		\$44,285,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.